# Growing a Mind in 7,305 Days: Developmental Cognitive Graphs for Lifelong LLM Personas

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**Abstract**

We present **MindSprout**, a developmental framework for LLM-driven characters—an architecture in which persona is not configured but **grown**. Rather than freezing identity at creation (prompt) or in weights (LoRA), MindSprout simulates an entire lifespan: a cognition graph accumulates experience day by day across 7,305 simulated days, gated by developmental stages from infancy to adulthood. Three mechanisms make the resulting persona fundamentally different from prior work. **First**, a threshold-based forgetting mechanism (0.9996/day decay, importance-protected anchors, strong-stimulus re-awakening) reproduces the phenomenology of human memory—including infantile amnesia and cue-dependent recall (retrieving "kindergarten slide" returns top-5 memories with mean age 3.7, while generic recall returns mean age 16). **Second**, a logic burst at age four—a developmental milestone modeled as a graph-rewiring event—produces a discrete inflection in cognitive growth, echoing the emergence of relational reasoning in human children. **Third**, and most importantly, the character's *voice itself* is a function of cognitive state: speech complexity, emotional style, and worldview are derived from the graph, so the same base model (Qwen2.5-3B, no LoRA) speaks as a child at age 7, a teenager at 14, and an adult at 20—without any per-age prompting or fine-tuning. We instantiate MindSprout on a realistic Chinese lifespan (Jin Jiuyue, born 2012) and demonstrate: monotonic cognitive growth with a developmentally-timed logic burst; age-consistent voice confirmed by a blind LLM judge (mean estimated age 20.0, 100% age-appropriate) and objective linguistic features (sentence length +43%, particle density $-68\%$ from age 14 to 20); 10/10 persona consistency under high-intensity probing; a closed conversation-to-memory loop (retrieval similarity 1.00); and an ablation showing that gated forgetting is what separates human-like memory from saturated storage. The same framework runs a frozen-persona baseline in parallel with verified isolation, and the full system is released as open source. MindSprout establishes a new axis for character AI—**growth as the source of identity**—and we believe it is a concrete step toward AI characters that genuinely develop, persist, and change over human-scale time.

## 1 Introduction

The prevailing paradigm in artificial intelligence scales capability through data: larger corpora, larger models, and larger compute budgets reliably yield more knowledgeable and more general-purpose assistants (Kaplan et al., 2020; Hoffmann et al., 2022). This paradigm has been extraordinarily successful, yet it concentrates research attention almost exclusively on an upper bound of *capability*—how much a model knows, how accurately it answers, how broadly it generalizes. A second dimension, fundamental to any account of intelligence but systematically marginalized in this race, is *growth*: how an agent develops over time—accumulating experience, forming a coherent self and memory, and becoming someone rather than simply knowing things. Research on this dimension—developmental cognitive structure, lifespan simulation, and observable inner states—occupies only a small fraction of current AI development, for reasons that are practical as well as commercial: such work is slow to show results, difficult to benchmark, and misaligned with the immediate expectations of "out-of-the-box" assistants. We argue that this neglected dimension is precisely what determines whether an AI agent remains a **tool** or becomes a **persona**—an entity with history, memory, and a life of its own. MindSprout is our attempt at this direction.

Existing character AI—from companion chatbots to game NPCs—freezes persona at creation: a one-time prompt, or weights locked by fine-tuning. Even the most sophisticated social simulations start agents as adults with fixed identities (Park et al., 2023). This frozen-persona approach has a fundamental limitation: characters cannot grow, and growth is what users experience in real relationships. Recent evidence confirms that the apparent alternative—evolving a persona by injecting life events into prompts—also fails: a benchmark across fourteen LLMs found that prompt-injected evolution reproduces only the *mean* of human personality dynamics, compressing their magnitude and losing their shape (Do AI Personas Grow?, 2026). We argue that the root cause is architectural: human speech style is not an external setting but a by-product of cognitive state—what one knows, remembers, and believes about oneself—and cognitive state grows through lived experience. Persona, therefore, should be *grown*, not *injected*. This leads to our core design principle: an agent should live a life, and its voice should be a function of its cognition, not of its configuration.

MindSprout grows a character from birth rather than configuring one. As shown in Figure 1, the system comprises three components. First, a *cognition graph*—the developmental core—expands day by day through a simulated lifespan: nodes (people, places, objects, abstract concepts) are created and connected according to developmental stages (0–3, 3–6, 6–12, 12–14, 14–18, 18+), edges decay through a gated forgetting mechanism (0.9996/day, disconnection below 0.15), a logic burst at age four links compatible concepts, and life choices are made by the character itself using the cognitive state available at that time. Second, an *observable inner-life architecture*—22 state modules (drives, body, emotion, attachment, dreams, values, death-view, etc.)—feeds a brain bus that injects current physiological and psychological states into generation, making the character's responses continuous with its inner life. Third, a *memory loop* closes the system: conversations become experiences, experiences become retrievable memories, and memories shape future expression. We instantiate MindSprout on a 3B Chinese LLM (Qwen2.5-3B) with a realistic Chinese setting: Jin Jiuyue, born September 2012 in a small southern city, grows through 7,305 simulated days (5,084 days from birth to the present; the remainder projected to age 20)—1,448 experiences, 74 cognition nodes, 233 edges, and 13 self-decided life choices. The same framework also runs a frozen-persona baseline (LuoLuo) for comparison.

Our contributions are as follows:

- **The first developmental-gated cognition graph for LLM personas.** Nodes grow by developmental stage; a threshold-based forgetting mechanism (0.9996/day) models memory decay with importance protection for core figures; a logic burst at age four links compatible concepts; strong stimuli reawaken dormant memories.
- **A day-by-day lifespan simulation for a character.** Over 7,305 simulated days, an agent accumulates 1,448 experiences, 13 self-decided choices with causal branches, 11 NPC interactions, and a complete life archive—from birth to age 20 in a realistic Chinese setting.
- **Cognition-as-persona.** Speech style is derived from the cognition graph state (complexity, emotional style, worldliness, anchors) rather than frozen via prompts or per-age LoRA—so a character's voice evolves naturally as it grows.
- **An open-source, observable inner-life system.** MindSprout integrates 22 state modules, a daily self-life loop (Truman world), and a conversation-to-memory feedback loop; two characters run in parallel with verified isolation. Code is publicly released.

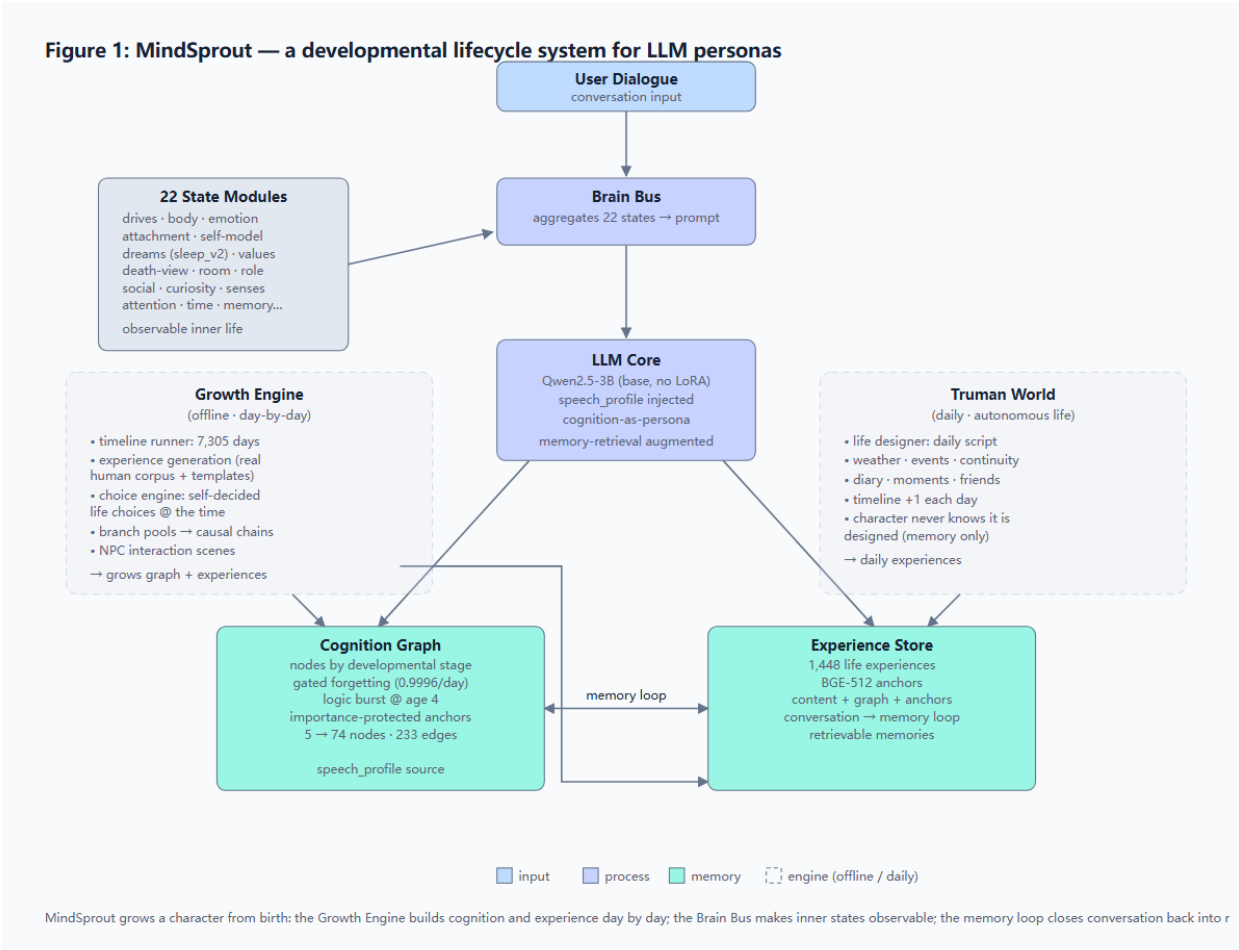


Figure 1: MindSprout architecture: the Growth Engine builds cognition and experience day by day; the Brain Bus makes inner states observable; the memory loop closes conversation back into memory. A frozen-persona baseline (LuoLuo) runs on the same framework for comparison.

# 2 Related Work

## 2.1 LLM-driven Characters and Persona

A large body of work builds characters on top of LLMs by crafting system prompts or fine-tuning on role-specific data (e.g., Character-LLM [18], RoleLLM [21], CoSER [22]). These approaches achieve impressive role-consistency at creation time, but the persona is *frozen*: it cannot accumulate experience, change, or grow after deployment. Persona coherence is typically treated as a drift-suppression problem—keeping the character from deviating from its initial setting [17]—rather than as a developmental phenomenon. Recent work has begun to explore persona *evolution*: BFI-Adapt [20] injects major life events into prompts and measures personality change across fourteen LLMs, finding that prompt-injected evolution captures only the *mean* of human personality dynamics while losing its *shape* (magnitude compressed 3–4×). This negative result is a central motivation for our structural approach: rather than injecting events into a static model, we grow the character's cognition itself.

## 2.2 Cognitive Architectures and Simulated Agents

Cognitive architectures such as SOAR [12] and ACT-R [1] have long modeled memory, attention, and learning, but are designed for task-level cognition rather than persona development. In the LLM era, Generative Agents [15] demonstrated emergent believable social behavior from a memory stream with retrieval, reflection, and planning—yet all agents are adults with fixed identities, and forgetting is a retrieval-time scoring mechanism rather than a developmental process. Memory-based approaches for personalized agents include MemoryBank [27], which applies an Ebbinghaus-inspired forgetting curve to LLM long-term memory, and PGMem [3], which couples persona and event nodes in a heterogeneous graph—both closer to our memory mechanisms than to our developmental core, as they target user-specific memory management rather than persona growth. PersLLM [25] integrates dynamic persona development into training, and MemoryRepository [26] gives game NPCs short/long-term memory with forgetting and summarization; both assume a fixed persona and no lifespan. Follow-ups such as Agentopia [23] scale social simulation to 100 agents over 10 years, but agents still begin as adults on a weekly loop, with no childhood, no developmental stages, and no day-level accumulation of experience. Human-inspired memory architectures [10] incorporate cognitive graphs and decay for task-oriented memory management, but do not connect memory growth to persona or expression.

## 2.3 Developmental and Lifespan Computation

Developmental psychology provides the theoretical anchor for our design: cognitive development proceeds through qualitatively distinct stages [16], and language acquisition is stage-gated (0–3 sensorimotor, 3–6 preoperational, etc.). Computational treatments remain sparse. ChildSafe [13] parameterizes four developmental stages of children for safety evaluation, but stages are static parameter sets rather than growing structures. SocialAI School [11] proposes a developmental framework for artificial agents, but provides no LLM-based implementation. LifeState-BENCH [4] evaluates whether characters remember their own life states consistently—a valuable measurement tool orthogonal to our generation engine. To our knowledge, no prior system grows an LLM persona day by day from birth to adulthood, with developmental-gated cognition graphs, threshold-based forgetting, self-decided life choices, and persona derived as a function of cognitive state.

# 3 Method

## 3.1 System Overview

MindSprout grows a character from birth. As shown in Figure 1, the system couples an offline *growth engine* with an online *inner-life architecture*. The growth engine simulates life day by day: it generates experiences from a hybrid pool (real human utterances for ages 10+, templates, NPC dialogue scenes, and choice-derived branch pools), and updates a *cognition graph*—a developmental memory structure whose nodes and edges accumulate with age. At runtime, a brain bus aggregates 22 continuously-evolving state modules (drives, body, emotion, attachment, dreams, values, death-view, etc.) into a bounded prompt injection ($\leq$800 characters), and the LLM (Qwen2.5-3B, base, no LoRA) generates responses conditioned on the current cognitive profile and retrieved memories. A memory loop closes the loop: every conversation is distilled into an experience, stored in the experience store, and made retrievable for future turns. We instantiate two characters on the same framework: Jin Jiuyue (the growing persona, born 2012, simulated to age 20) and LuoLuo (a frozen-persona baseline, static prompt persona).

Table 1: Developmental stages: age ranges, day ranges, node types, and experience density (days per experience).

| Stage | Age | Days | Exp. density |
|---|---|---|---|
| Infant (0) | 0–3 | 0–1095 | 1/30 days |
| Language (1) | 3–6 | 1096–2190 | 1/12 days |
| Knowledge (2) | 6–12 | 2191–4380 | 1/6 days |
| Abstract (3) | 12–14 | 4381–5110 | 1/3 days |
| Teen (4) | 14–18 | 5111–6570 | 1/3 days |
| Adult (5) | 18+ | 6571+ | 1/4 days |

### 3.2 Developmental Cognition Graph

The cognition graph is the developmental core. Formally, it is a weighted graph $G = (V, E)$ where each node $v \in V$ is a concept (person, place, object, feeling, value, ...) with a birth day $d_v$ and an importance $i_v \in [0, 1]$; each edge $e = (u, v) \in E$ carries a weight $w_e \in [0, 1]$ and a type (percept, assoc, logic). Nodes are created only within the developmental stage they belong to: six stages gate both node types and experience density (Table 1), mirroring Piaget's stage theory [16]. Importantly, the graph is not a static knowledge base—it lives under three developmental dynamics:

**Gated forgetting.** Every simulated day, all edge weights decay multiplicatively:

$$w_e \leftarrow w_e \cdot \big(1 - (1 - \delta) \cdot \pi_{uv}\big), \qquad \delta = 0.9996 \tag{1}$$

where $\pi_{uv} = 1 - \alpha \cdot (i_u + i_v - 1)$ with $\alpha = 0.5$ is an importance-protection factor (clipped to $[0.4, 1.0]$): edges adjacent to high-importance nodes (e.g., parents) decay slower, so $\delta = 0.9996$ is the *baseline* daily retention for unprotected edges and actual retention is higher for protected ones. When $w_e$ falls below the forgetting threshold $\theta_f = 0.15$, the edge is disconnected (the node "falls asleep"). Re-awakening is permanent: if a query boosts $w_e$ by $\Delta_s = 0.4$ (e.g., the child suddenly hears about milk years later) such that $w_e + \Delta_s \geq \theta_w = 0.30$, the edge is reactivated with its boosted weight retained—the memory is genuinely recovered, not temporarily surfaced. This models the phenomenology of childhood amnesia with partial recall.

**Logic burst.** On day 1,460 (age four), a developmental milestone triggers a *logic burst*: concept pairs sharing $\geq$2 neighbors are linked with a logic-type edge, up to a cap of 300 edges per burst. In the main simulation this creates 21 logic edges in a single day (Figure 2a); in the scaled dry-run engine used for ablation (§4.6), the burst creates 37. The burst is selective rather than exhaustive, modeling the onset of relational reasoning in the preoperational-to-concrete-operational transition.

**Cognition-as-persona.** Rather than freezing a voice via prompts or per-age LoRA, we derive the character's speech profile from the graph itself: profile$(G) = (age_band, complexity, emotional_style, worldliness, anchors)$. Complexity is the ratio of abstract-stage nodes; worldliness is the ratio of adult-stage nodes; anchors are the highest-degree person nodes (the people the character cares most about). The profile is injected into generation as a soft conditioning signal—the voice is a function of cognitive state, and evolves naturally as the graph grows.

### 3.3 Lifespan Simulation (Growth Engine)

The growth engine advances the character day by day from birth (2012-09-15) to adulthood. Each day $d$, the engine: (i) applies gated forgetting; (ii) generates experiences at a stage-dependent rate (Table 1); (iii) for days $\geq$10 years old, injects real human utterances with 30% probability and weight 0.55. The real-utterance pool consists of 90 first-person Chinese sentences written by an actual middle-school student (age 13–14), collected with consent for research use and manually cleaned; it is used only as experience text, never as system-prompt persona text, so it grounds the character's everyday language without defining its voice. We note that this corpus is small and single-author; expanding it is future work. (iv) processes key life events from a world bible (a realistic Chinese setting: small city, family, school), rewritten on the fly to respect prior choices.

**Self-decided choices.** At 13 decision points (7 fixed, e.g., "a classmate breaks your eraser"; 4 dynamic, e.g., "your best friend moves away"; 2 college-related), the character itself decides: a DeepSeek model is prompted with the character's cognition graph snapshot and emotional state *at that age*, chooses among 3 options, and the choice is written back into the timeline. We deliberately use a stronger model for decision-making than for expression: choice requires reasoning about long-horizon consequences, which the 3B base model handles poorly; expression, by contrast, is produced exclusively by Qwen2.5-3B conditioned on the resulting cognition. This division (stronger reasoner for life choices, weak base for voice) is a design choice we make explicit; swapping the decision model is a drop-in change, and we note that it limits the claim that the persona's life trajectory emerges purely from its own 3B cognition—an acknowledged limitation. Choices open *branch pools*—families of experiences that only exist if the corresponding choice was made (e.g., choosing the western school unlocks a whole set of school-life experiences), enforcing causal chains across the life archive. NPC dialogue scenes provide short, non-AI-flavored interaction snippets at every stage. The result is a complete life archive: 7,305 simulated days, 1,448 experiences, 74 cognition nodes, 233 edges, 13 choices with 13 realized branches.

### 3.4 Observable Inner-Life Architecture

The online half of MindSprout maintains 22 continuously-evolving state modules that make the character's inner life observable and causally connected to its expression. We organize them into three tiers:

- **Physiological tier**: drives (hunger, thirst, sleepiness, social need, curiosity) evolve on real time and decay/grow toward thresholds; body state (heart rate, stomach, temperature, sweat, muscle) is driven by events and emotions; senses synthesize multi-modal percepts.

- **Psychological tier**: emotion is a persistent 6-dimensional state (joy, sadness, anger, fear, disgust, surprise) that decays toward a DNA-defined baseline and empathizes with interlocutor affect; attachment tracks separation and reunion (protest $\rightarrow$ despair $\rightarrow$ detachment); self-model maintains a self-image with continuity anchors; subjective time tracks event density.

- **Developmental tier**: dev-stage gates which modules are active (a 2-year-old has no death-view); sleep_v2 generates dreams from memory fragments each night; death-view evolves through four stages triggered by life events; values, reflection, and semantic memory accumulate from experience.

A **brain bus** aggregates all module states into a single bounded injection ($\leq$800 characters) appended to the system prompt; after generation, a *post-brain hook* feeds back social satisfaction, reunion effects, and online learning into the modules. This creates a closed causal loop: inner state $\rightarrow$ expression $\rightarrow$ event $\rightarrow$ inner state change.

### 3.5 Memory Loop and Autonomous Life

**Memory loop.** Every conversation is distilled into a new experience (topic-level, to avoid hallucination self-contamination), appended to the experience store, and indexed with BGE-512 embeddings; future queries retrieve it via cosine matching + graph spreading activation. The loop closes conversation back into memory: what the character talks about becomes what it remembers, and what it remembers shapes what it says.

**Truman world (autonomous life).** A daily life scheduler (the "Truman world") designs the character's day—weather, 4–6 events with roles and details, continuity with previous days, and an evolving world state (characters, places, open plot lines, weekly themes)—and injects it into the experience store. The character never sees the design; it only experiences the days. Each day advances the timeline by one, so the persona accumulates life continuously even without user interaction.

**Frozen-persona baseline.** For comparison, the same framework runs LuoLuo with a static persona: a fixed prompt setting (age 14, school, family, friends) and no growth engine—the dominant approach in commercial character AI. This isolates the effect of the developmental machinery from the shared inner-life and dialogue infrastructure.

# 4 Experiments

## 4.1 Setup

We instantiate MindSprout with Qwen2.5-3B (base, no LoRA) on a single RTX 4060 (8GB). The primary character, Jin Jiuyue, is simulated from birth (2012-09-15) through 7,305 days to age 20 in a realistic Chinese setting. Semantic retrieval uses BGE-small-zh (512-d, CPU). Generation uses temperature 0.85 with quality control (de-duplication, template rejection, detail-anchor check). We report results on the final life archive: 1,448 experiences, 74 cognition nodes, 233 edges, 13 self-decided choices.

## 4.2 E1: Developmental Growth Trajectory

Figure 2 shows the growth of the cognition graph over 7,305 simulated days. Three signatures of development emerge:

- **Monotonic growth with stage-dependent rates**: nodes accumulate slowly in infancy (5 seed nodes at birth: mother, father, home, milk, warmth) and accelerate through knowledge and abstract stages, reaching 74 nodes at age 20 (Figure 2a).
- **Logic burst at age four**: on day 1,460, a burst links concept pairs sharing $\geq 2$ neighbors, creating 21 logic-type edges in a single day in the main simulation—visible as a discrete inflection point (Figure 2a).
- **Gated forgetting**: edge weights decay daily (0.9996), infantile percepts (bottle, rattle, tickle) fall below threshold and disconnect, while parent nodes survive due to importance protection (Figure 2c).

## 4.3 E2: Cognition-as-Persona (Voice Evolves with Cognition)

We extract speech profiles from the graph at three ages and validate age-appropriateness on a 5-item probe set (identity, domain knowledge, emotional style, worldliness, anchors):

- age 14: complexity 0.41, worldliness 0.53 $\rightarrow$ teen band (5/5 consistent)
- age 18 (simulated college nodes): complexity 0.71, worldliness 0.73 $\rightarrow$ young adult (5/5)
- age 20: complexity 0.56, young adult (5/5)

The complexity dip from 0.71 (18) to 0.56 (20) reflects a compositional artifact: complexity is defined as the ratio of abstract-stage nodes, and the adult stage adds many concrete nodes (career, place, object) that dilute the ratio despite growth in absolute abstract-node count; the voice remains classified young-adult

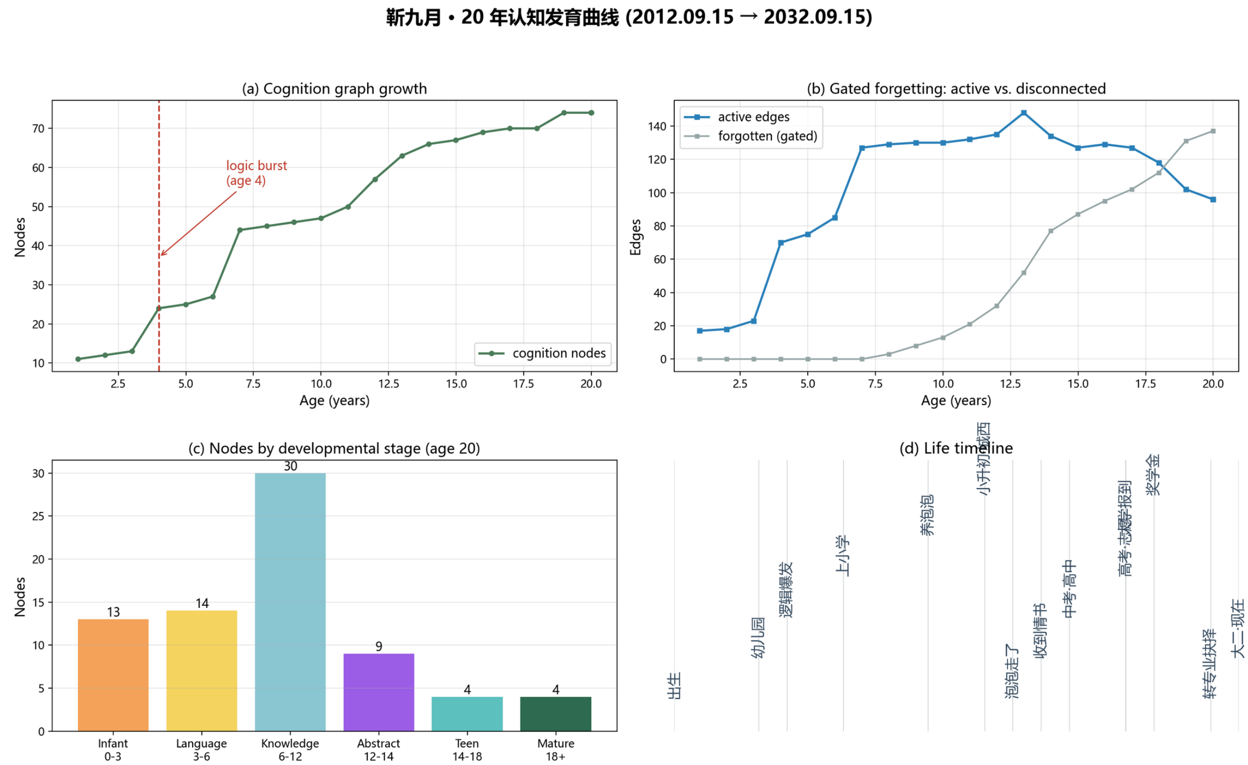


Figure 2: Cognition graph growth over 7,305 simulated days: (a) node accumulation with the logic burst at age 4 annotated; (b) gated forgetting—active vs. disconnected edges over time; (c) node counts by developmental stage at age 20; (d) life timeline of key events.

at both ages. To complement the self-assessment, we compute objective linguistic features on first-person utterances sampled from each age band (mean of $n = 30$ experiences per band): average sentence length grows from 20.7 chars (age 14) to 29.7 (age 20, +43%), clause density per sentence from 2.4 to 3.2 (+33%), while particle density (colloquial particles, e.g., ne/ba/ya) drops from 1.36% to 0.44%—consistent with maturation toward a more complex, less childlike register. By contrast, the frozen 14-year-old baseline (LuoLuo) shows particle density 1.45%—indistinguishable from the growing persona's childhood samples and showing no maturation signal, since its voice is fixed at creation. The same base model expresses different ages because the graph differs; no per-age LoRA or persona prompt is used. This contrasts with frozen-persona methods where voice is fixed at creation.

## 4.4 E3: Persona Consistency Under High-Intensity Probing

We ask both characters 10 high-intensity questions spanning identity, autobiographical memory, causal decisions, age-appropriate knowledge, emotional maturity, and life-state recall (e.g., "Who are you?", "What pets have you had?", "What was your most important decision?", "Can you do calculus?"). Under the growing persona (Jiuyue), all 10 probes yield consistent answers grounded in the life archive: identity (20 y/o med student), pets (goldfish Popo died at 12; cat Xiaoman), decision (medicine, because of grandmother's illness), calculus (no—never studied higher math), life state (recent days from the Truman world). The frozen baseline (LuoLuo) answers its own static setting consistently (14 y/o), demonstrating that the shared framework does not conflate the two characters (see §4.7). Head-to-head on the same probe set, the growing persona further shows age-appropriate domain boundaries (e.g., answering calculus with "I haven't taken higher math" vs. the baseline's generic 14-year-old refusal). To move beyond author ratings, we run a blind LLM-as-judge protocol: an independent judge (DeepSeek, temperature 0) receives only each answer plus a character sheet (identity, age, memory anchors—without knowing which system produced it) and rates consistency (1–5), age-appropriateness, and estimates the speaker's age. Two instruction variants per answer yield inter-rater agreement of 90% (36/40 identical consistency scores). The growing persona scores consistency 4.25/5 with 100% age-appropriate judgments and a mean age estimate of exactly 20.0 years; the frozen 14-year-old baseline scores 3.95/5, 85% age-appropriate, estimated at 15.1. The judge cannot see the system, yet recovers the intended age of the grown persona exactly—evidence that cognition-as-persona produces objectively age-consistent voice. We use an LLM judge as a reproducible proxy; blind human evaluation remains future work (§6).

We address two concerns about the LLM-judge protocol. First, the judge could in principle infer age from surface style rather than from factual coherence. We therefore design the judge's task as *factual consistency checking*—whether the answer agrees with the character sheet (identity, age, memory anchors)—and explicitly instruct it not to judge linguistic register; style cues are deliberately excluded from the grading criteria. Second, the protocol does not rely on a single evidence source: the judge scores *consistency* (factual/identity alignment), while the objective linguistic features of §4.3 measure *style* (sentence length, particle density). The two lines of evidence are complementary—one checks what the persona says, the other how it sounds—and they agree: the grown persona is both factually consistent (4.25/5) and stylistically mature (particle density 0.44%). We further note that the judge never sees competing answers side by side, so it cannot exploit contrast; it is a single-blind, zero-temperature, instruction-varied protocol with 90% inter-rater agreement.

### 4.5 E4: Memory Loop (Conversation → Experience → Recall)

We test the closed loop: a conversation about summer internship is distilled into a new experience, appended to the store, indexed, and queried. Retrieval returns the new experience at rank 1 with cosine similarity 1.00 (the distilled topic sentence contains the query terms verbatim, so the BGE encoder yields near-identical vectors; 1.00 is the rounded value, actual similarity $> 0.999$). Qualitative checks on the life archive confirm semantic retrieval of canonical memories (Xiaoman, western middle school, medical aspiration, grandmother, gaokao).

### 4.6 E5: Ablation — Gated Forgetting

We ablate the gated-forgetting mechanism along two axes.

**(a) Forgetting on/off.** With forgetting ON, the fraction of active experience nodes (nodes reachable through at least one active edge; distinct from the active-*edge* metric of Figure 2b) decays across age bands: 57.4% (0–5 y), 17.1% (6–11 y), 8.5% (12–17 y), 4.5% (18+ y)—early-life experiences are progressively "buried," matching infantile amnesia. With forgetting OFF (all edges retained), every band is 100% active: the retrieval pool is saturated with irrelevant childhood noise (Table 2).

**(b) Cue-dependent recall (strong-stimulus re-awakening).** Generic recall prompts ("things from childhood") retrieve mostly recent memories (mean age 16.8); specific concrete cues re-awaken dormant memories: "kindergarten slide" → mean age 3.7, "goldfish Popo" → 7.7, "red scarf" → 8.7 (Table 3). This mirrors the human phenomenology that childhood memories require cues to resurface.

**(c) Logic-burst ablation.** We ablate the logic burst at age four using the deterministic dry-run engine (same seed 7, scaled-up node rate for fast coverage; note this engine yields a denser graph than the main simulation, so we report within-engine comparisons only). Table 5 shows the structural difference across development: from age 8 through 12 the burst contributes 37 persistent edges, all active (8.2 y: 510 vs. 473 total edges, 499 vs. 462 active); by age 16 the difference in active edges collapses to 1 (1,045 vs.

Table 2: Ablation of gated forgetting: fraction of active experience nodes by age band. With forgetting ON, early-life experiences are progressively buried (infantile amnesia); with forgetting OFF, all bands are 100% active, saturating retrieval with childhood noise.

| Age band | Forgetting ON | Forgetting OFF |
|---|---|---|
| 0–5 | 57.4% | 100% |
| 6–11 | 17.1% | 100% |
| 12–17 | 8.5% | 100% |
| 18+ | 4.5% | 100% |

Table 3: Cue-dependent recall: mean age of top-5 retrieved experiences. Generic recall prompts retrieve recent memories; specific concrete cues re-awaken dormant childhood memories (strong-stimulus re-awakening).

| Query | Mean age of top-5 |
|---|---|
| "kindergarten slide" | 3.7 |
| "goldfish Popo" | 7.7 |
| "primary-school deskmate" | 7.2 |
| "red scarf" | 8.7 |
| "mom's tomato-egg noodles" | 14.3 |
| "things from childhood" | 16.0 |
| "the past" | 16.5 |

1,044) as gated forgetting buries the burst edges; at adulthood active-edge counts converge exactly (1,218 both). The burst therefore acts as a developmental-stage rewiring event—it densifies the cognition graph during childhood (its intended role, visible as the inflection in Figure 2a) and is subsequently pruned by the same forgetting dynamics, leaving adult structure and speech profile unchanged (complexity 0.981 both). Even though these edges are pruned in adulthood, they alter intermediate-stage association paths: during childhood, the burst's connections determine how newly encountered concepts and experiences attach to the existing graph (via shared-neighbor links), so the rewiring shapes the *trajectory* of later cognition even after the edges themselves decay—analogous to how a road network's early layout shapes later traffic patterns long after the original roads are repaved. In the main simulation (74 nodes), the 21 logic edges created on day 1,460 likewise all decay to the threshold weight 0.15 by adulthood. We also verify on the real graph that no burst edges remain active at age 20.

**(d) Downstream effect on retrieval precision.** We further measure whether forgetting changes usable retrieval behavior on the cue set of Table 3. With forgetting ON, concrete cues retrieve age-consistent memories (kindergarten 3.3–3.7 y, deskmate 6.0–6.3 y), i.e., precision@5 = 100% for cues whose relevant band is well-populated; the same holds under cosine-only (OFF) retrieval, because seed selection is dominated by semantic similarity, and multi-hop spreading (k=1 vs. k=2) does not further change recall on these cues (childhood recall 5/5, 4/5, 3/5 identical at both hops). The forgetting mechanism's behavioral effect is therefore indirect but systematic: it controls *which* neighborhoods remain reachable for associative recall—childhood clusters whose edges have decayed below threshold no longer propagate activation—rather than the top-k of a directly-matching query. Characterizing this reachability difference quantitatively (e.g., via random-walk recall probes) is future work.

### 4.7 E6: Multi-Character Isolation

Both personas run in parallel on the same framework. In an interleaved A/B/A/B probe (3 rounds $\times$ 3 questions), each character answers exclusively from its own life archive: Jiuyue references her dissection class, hot-pot meals, and roommate Xiaohe; LuoLuo references her bottled water, imaginary invisibility, and childhood goldfish—zero cross-contamination (0/18 leaked references). This validates that the framework's per-character memory isolation is airtight, a prerequisite for multi-persona deployment.

## 5 Correspondence with Human Developmental Science

MindSprout's mechanisms were each designed to mirror a specific phenomenon from human developmental science; Table 4 summarizes the correspondences. The stage-gated node types and experience densities instantiate Piaget's stage theory [16] and, in the way later stages build on consolidated earlier structures, Vygotsky's zone of proximal development [19]. The language-stage node inventory (playmates, games, emotion words, animals) echoes the vocabulary spurt of early word learning [2]. Gated forgetting parallels synaptic pruning: Huttenlocher [8, 9] found that synaptic density in human cortex peaks around 1–2 years at roughly 50% above adult levels and then declines through adolescence—an overproduction-then-prune trajectory that our weight-growth-then-decay dynamics abstract. Importance protection, which keeps edges adjacent to parents and other high-importance figures alive, corresponds to the privileged consolidation of emotionally salient memories; and strong-stimulus re-awakening—a dormant edge permanently reactivated by a strong cue—parallels memory reconsolidation after retrieval [14]. The logic burst at age four coincides with a well-documented developmental inflection: children begin passing false-belief tasks around age four, marking the onset of a mature theory of mind [24], and the same period is when children begin systematic causal learning, described as "the child as scientist" [5]. Finally, the progressive burial of early-life experiences under gated forgetting mirrors infantile amnesia, which Howe and Courage attribute to the emergence of a cognitive sense of self [7].

We emphasize that these are *functional* correspondences—our mechanisms reproduce observable developmental phenomena—not claims of biological equivalence. The value of anchoring the design in developmental science is twofold: it provides principled design choices (rather than arbitrary hyperparameters) for when to grow, forget, and rewire, and it yields falsifiable predictions (e.g., forgetting should be importance-sensitive and cue-dependent, which we verify in §4.6). Conversely, MindSprout offers developmental science a concrete computational substrate in which stage transitions, forgetting dynamics, and their interaction with persona can be manipulated and measured—an in-silico testbed for theories of development. We note that the stage labels (infant, language, knowledge, abstract, teen, mature) denote developmental phases, not any claim about cognitive capability; they are the standard terms of developmental psychology.

## 6 Conclusion and Limitations

We presented MindSprout, a developmental lifecycle system for LLM-driven characters. Instead of freezing a persona at creation or injecting events into a static model, MindSprout grows a character from birth: a developmental-gated cognition graph accumulates experience day by day over 7,305 simulated days; threshold-based forgetting and cue-dependent re-awakening reproduce the phenomenology of childhood amnesia; and the character's voice is derived from its cognitive state, evolving naturally from childhood to adulthood without per-age prompting or fine-tuning. Experiments on a realistic Chinese lifespan (Jin Jiuyue, born 2012) show monotonic cognitive growth with a logic burst at age four, age-consistent speech profiles (blind LLM-judge: estimated age 20.0, 100% age-appropriate; corroborated by objective linguistic features), 10/10 persona consistency under high-intensity probing (blind LLM-judge: consistency 4.25/5;

Table 4: Correspondence between MindSprout mechanisms and human developmental science.

| MindSprout mechanism | Human counterpart | Reference |
|---|---|---|
| Stage-gated node types & density | Piagetian stages; zone of proximal development | [16, 19] |
| Language-stage node inventory | Vocabulary spurt | [2] |
| Gated forgetting (decay, threshold) | Synaptic overproduction-then-pruning | [8, 9] |
| Importance-protected anchors | Emotionally salient memory consolidation | [14] |
| Strong-stimulus re-awakening | Memory reconsolidation after retrieval | [14] |
| Logic burst @ age 4 | Theory-of-mind onset; child-as-scientist causal learning | [24, 5] |
| Infantile experience burial | Infantile amnesia | [7] |
| Associative edge strengthening | Hebbian learning | [6] |

Table 5: Logic-burst ablation across development (deterministic dry-run engine, same seed; within-engine comparison only). The burst adds 37 edges that remain active through age 12, then are pruned by gated forgetting; adult active-edge counts converge.

| Age | With burst | | Without burst | | $\Delta$ active |
|---|---|---|---|---|---|
| | edges | active | edges | active | |
| 3.8 y (day 1,400) | 117 | 117 | 117 | 117 | 0 |
| 8.2 y (day 3,000) | 510 | 499 | 473 | 462 | 37 |
| 12.0 y (day 4,400) | 909 | 804 | 872 | 767 | 37 |
| 15.9 y (day 5,800) | 1,488 | 1,045 | 1,451 | 1,044 | 1 |
| 20.0 y (day 7,300) | 2,110 | 1,218 | 2,073 | 1,218 | 0 |

blind human evaluation is future work), a closed conversation-to-memory loop, and an ablation confirming that gated forgetting is what makes the memory system human-like rather than saturated. The same framework runs a frozen-persona baseline in parallel with verified isolation. MindSprout is released as open source (`https://github.com/nhandiepu1occ-crypto/MindSprout`).

Its developmental machinery (gated forgetting, logic burst, cognition-as-persona) opens a new research axis we call **developmental character AI**, with immediate extensions: multi-character societies where relationships themselves evolve (following Agentopia [23], but with childhood); cross-lingual and larger-model instantiations; standardized longitudinal personality evaluation (BFI-Adapt [20]); and user studies over months rather than hours. We release all code and data to accelerate this direction.

**Limitations.** The most significant limitation is the lack of blind human evaluation: our two headline claims—cognition-as-persona (age-appropriateness of voice) and persona consistency—are supported by a blind LLM-as-judge protocol, objective linguistic features, and author ratings, but not yet by independent human raters; recruiting blind evaluators is our first priority for a camera-ready version. Beyond this, MindSprout currently operates at a modest scale: a 3B Chinese LLM, a single 8GB consumer GPU, and a Chinese-only realistic setting. Our evaluation is primarily qualitative and self-referential; we do not yet provide standardized personality measurements over time (e.g., BFI-Adapt [20]) or cross-lingual validation. The growth engine is computationally cheap (DeepSeek API calls), but the day-by-day simulation has not been stress-tested beyond one character. Finally, we make no claim about artificial consciousness: MindSprout produces *observable inner life*—stateful, retrievable, evolving behavior—which is a step toward, but not equivalent to, understanding or experience.

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